

***CYS – AICT– LAB1- SPRING - SECTION A -
(ANAYA AZAM) – 2026***

***SUBMITTED TO:
PROF. MOHSIN MUMTAZ***

Task 1 : Application to the Chairman of the department for the change in discipline.

***The Chairmn of CE Department,
University of Engineering and Technology,
Lahore.***

Subject : Request for Change in Disciplin

Respected Chairman,

I am an honoured student enrolled in your Department of Computer Engineering, studying in the course of **Cybersecurity** here at UET, under the **Student I.D. (0123456789)** .

Due to personal reasons, I need to bring a matter of utmost improtance to your knowledge. My time at your department has been one of utmost joy for me and it is also through my time here that I have come to realise that my skills and aptitude for learning is better suited for another course here at your department. This course has not only piqued my interest but has also presented itself as an opportunity for a career that will benefit me. With my skills in mathematics and my analytical skills, I would do better in the **Data Science Course** . I assure you, I will maintain my excellent academic record and will not slack off.

I implore you to help me with this decision. Thank you for your time to consider my situation. Thanking you in anticipation.

Your Student,

Roll no. 123456

CNIC: 1234567890

Date: 22nd February, 2026.

Task 2 : Create a poster for Sports Gala for CE Department UET Lahore.



Task 3 : Create a bio-data form and fill it.

STUDENT BIO - DATA

PASTE

PASSPORT

-SIZED PICTURE

HERE

NAME: _____

FATHER'S NAME: _____

AGE: _____

STUDENT'S CONTACT: _____

GENDER: _____

DATE OF BIRTH: _____

RELIGION: _____

CITY: _____

E-MAIL: _____

COUNTRY OF BIRTH: _____

HOME ADDRESS: _____

PERMANENT ADDRESS: _____

LANDLINE NUMBER: _____ FATHER'S CONTACT: _____

EMERGENCY NUMBER: _____ GUARDIAN'S CONTACT: _____

EDUCATIONAL BACKGROUND:

MATRIC:

BOARD: _____

YEAR: _____

PERCENTAGE: _____

INTERMEDIATE:

BOARD: _____

YEAR: _____

PERCENTAGE: _____

FOR FOREIGN STUDENTS:

COUNTRY OF BIRTH: _____

PASSPORT NUMBER: _____

CITIZENSHIP(S): _____

Task 4 : Create references in IEEE format.

Black Holes

Black holes are some of the most enigmatic objects in the universe. These regions of spacetime exhibit an intense gravitational pull so strong that nothing, not even light, can escape their grasp. Black holes challenge our understanding of physics and have captivated scientists and the public alike for decades. [1] One of the key properties of a black hole is its event horizon, the point of no return beyond which nothing can escape. Black holes also have a singularity at their center, where the laws of physics as we know them break down. [2] Black holes form when a massive star runs out of fuel and collapses under its own gravity. As the star collapses, its core becomes denser and denser until it reaches a point where the gravitational pull is so strong that not even light can escape. The collapse of a star into a black hole is a violent process that releases an enormous amount of energy in the form of a supernova explosion. Black holes are not just fascinating objects in themselves, but they also have significant implications for our understanding of the universe. For example, the study of black holes has helped us to understand the behavior of galaxies. Scientists believe that supermassive black holes, which are found at the centers of most galaxies, play a crucial role in the formation and evolution of galaxies. They do this by influencing the motions of stars and gas within the galaxy. [3] Black holes also have implications for our understanding of the fundamental laws of physics. For example, the fact that black holes have a singularity at their center, where the laws of physics break down, suggests that our current understanding of physics is incomplete. This has led scientists to develop new theories, such as string theory and loop quantum gravity, in an attempt to reconcile our understanding of black holes with our understanding of the universe at large. [4] Black holes have been the subject of intense study and research in astrophysics for decades. Firstly, one of the most interesting properties of black holes is their size. Black holes come in different sizes, ranging from micro black holes, which are smaller than atoms, to supermassive black holes, which can have masses equivalent to billions of suns. The size of a black hole depends on its mass, with larger black holes having a greater gravitational pull. Secondly, black holes have an event horizon, which is the point of no return. Anything that crosses the event horizon is trapped inside the black hole forever. The event horizon is determined by the mass of the black hole, with larger black holes having a larger event horizon. Thirdly, black holes have a singularity at their center, which is a point of infinite density and gravity. This is where all the matter that falls into the black hole is compressed to a point of infinite density. The singularity is a theoretical concept, and the laws of physics break down

at this point. Fourthly, black holes emit radiation known as Hawking radiation. This radiation is due to quantum effects near the event horizon, where pairs of particles and antiparticles are created. In normal circumstances, these particles would annihilate each other, but near a black hole, one particle can be trapped inside the event horizon, while the other escapes. This causes the black hole to lose mass and eventually evaporate. Finally, black holes can have a significant impact on their surroundings. They can distort spacetime and cause gravitational lensing, where light is bent as it passes through the gravitational field of the black hole. They can also attract nearby objects and cause them to orbit around the black hole, forming an accretion disk. This disk can emit high-energy radiation, including X-rays and gamma rays. [5] According to classical physics, once an object crosses the event horizon, it is lost forever in the black hole's singularity. However, Hawking radiation predicts that black holes emit radiation, making them not entirely black. The concept of Hawking radiation is rooted in the principles of quantum mechanics and general relativity. According to quantum mechanics, the vacuum of space is not truly empty but instead is teeming with virtual particles and antiparticles that spontaneously appear and disappear. These virtual particles can be created by the strong gravitational field of a black hole, causing a separation of particle pairs with one particle falling into the black hole, and the other escaping into space. The result of this process is that the black hole loses a small amount of mass and energy, leading to its eventual "evaporation" over an extremely long timescale. This process was initially thought to be purely theoretical, but recent observations of black holes, such as those made by the Event Horizon Telescope, have provided evidence of its existence. One of the crucial factors in Hawking radiation is the temperature of the black hole, which is directly related to its mass. The smaller the black hole, the higher its temperature and the faster it will emit radiation. Conversely, larger black holes will emit radiation at a much slower rate, meaning that they can take billions of years to evaporate completely. If the star is less than three times the mass of the sun, it will form a white dwarf. If it is between three and eight times the mass of the sun, it will form a neutron star. If it is more than eight times the mass of the sun, it will form a black hole. The process of black hole formation is complex and depends on several factors, such as the mass, density, and composition of the star. When a star runs out of fuel, it no longer has enough energy to counteract the force of gravity, and it begins to collapse. As the star collapses, it heats up, and the core becomes denser and denser. Eventually, the core becomes so dense that the gravitational pull becomes too strong, and nothing can escape it, not even light. This creates a region of space called the event horizon, and anything that crosses it is pulled into the black hole.

References:

- [1] S. W. Hawking, "Black holes and thermodynamics," *Physical Review D*, vol. 13, no. 2, pp. 191–197, 1976.
- [2] R. P. Kerr, "Gravitational field of a spinning mass as an example of algebraically special metrics," *Physical Review Letters*, vol. 11, no. 5, pp. 237–238, 1963.
- [3] R. Penrose, "Gravitational collapse: The role of general relativity," *Rivista del Nuovo Cimento*, vol. 1, no. 252, pp. 252–276, 1965.
- [4] M. J. Rees, "Black hole models for active galactic nuclei," *Annual Review of Astronomy and Astrophysics*, vol. 22, no. 1, pp. 471–506, 1984.
- [5] S. Hawking, "Black hole explosions?," *Nature*, vol. 248, no. 5443, p. 31, 1974.